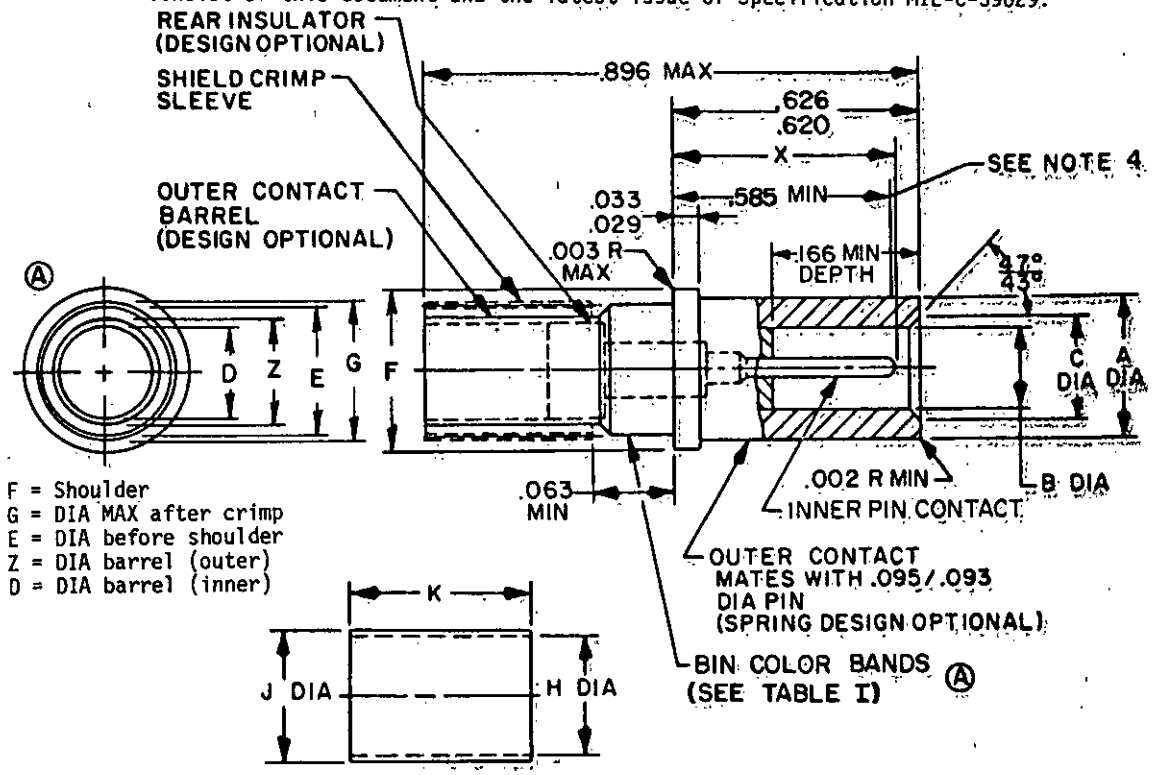
MILITARY SPECIFICATION SHEET

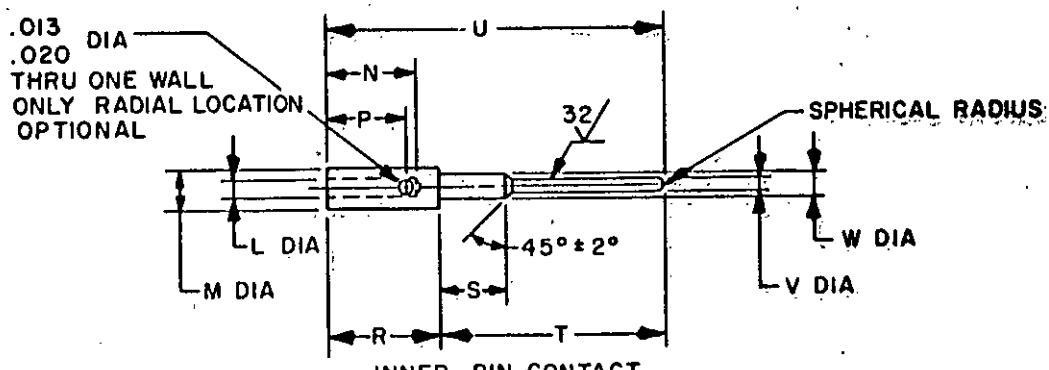
CONTACTS, ELECTRICAL CONNECTOR, SOCKET, CRIMP REMOVABLE, SHIELDED, SIZE 12 (FOR MIL-C-38999 SERIES I, II, III AND IV CONNECTORS)

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the contacts described herein shall consist of this document and the latest issue of Specification MIL-C-39029.



SHIELD CRIMP SLEEVE



INNER PIN CONTACT

Ⓐ Denotes Changes

FIGURE 1. Socket contacts.

FSC 5935

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BIN code	A Dia.	B Dia.	C Dia.	D Dia. Min.	E Dia.	F Dia.	G Dia. Max.	H Dia. Min.	J Dia. Max.	K	L Dia. Min.	M Dia. Max.
416	.161	.100 .097	.123 .118	.090	.151 .148	.182 .179	.156	.127	.169	.125 .115	.0225	.052
417				.108				.144			.0225	
418				.108				.144			.0355	
419				.090				.127			.0355	
420				.090				.127	.169		.0225	
421				.117				.156	.174		.0270	
422				.090				.144	.169		.0225	
423	.161	.100 .097	.123 .118	.108	.151 .148	.182 .179	.156	.156	.174	.125 .115	.0355	.052

BIN code	N Min.	P	R	S	T	U REF.	V Dia.	W Dia.	X	Z Dia. Max.
416	.112	.103 .096	.146 .140	.039 .033	.222 .219	.3635	.0205 .0195	.035 .033	.613 .603	.110
417										.127
418										.127
419										.110
420										.110
421										.136
422										.110
423	.112	.103 .096	.146 .140	.039 .033	.222 .219	.3635	.0205 .0195	.035 .033	.613 .603	.127

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.002	.05	.0355	.902	.100	2.54	.125	3.18	.158	4.01	.3635	9.233
.003	.08	.039	.99	.103	2.62	.127	3.23	.161	4.09	.585	14.86
.0195	.495	.052	1.32	.108	2.74	.136	3.45	.166	4.22	.603	15.32
.0205	.521	.063	1.60	.110	2.79	.140	3.56	.169	4.29	.613	15.57
.0225	.572	.090	2.29	.112	2.84	.144	3.66	.174	4.42	.620	15.75
.027	.69	.093	2.36	.115	2.92	.146	3.71	.179	4.55	.626	15.90
.029	.74	.095	2.41	.117	2.97	.148	3.76	.182	4.62	.896	22.76
.033	.84	.096	2.44	.118	3.00	.151	3.84	.219	5.56		
.035	.89	.097	2.46	.123	3.12	.156	3.96	.222	5.64		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
3. Dimensions shown apply after plating.
4. Point at which a square ended pin of the same basic diameter as the mating contact first engages the outer contact spring. Provision for clearance hole shall be provided.
5. Crimp deformation: The maximum diameter over the crimped portion of the shield crimp sleeve shall not exceed G diameter.

FIGURE 1. Socket contacts - Continued.

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REQUIREMENTS:

Contacts shall comply with the reliability assurance provisions of MIL-STD-790 as specified in MIL-C-38999.

Dimensions, design characteristics, and configuration: See figure 1 and table I.

Mating contacts: MIL-C-39029/28.

Tools: See table II.

TABLE I. Design characteristics.

BIN code	Color bands			Cable accommodated	Contact cavity size	Type	Class
	1st	2nd	3rd				
416	Yellow	Brown	Blue	1/2/ M17/119-RG174 M17/113-RG316 M17/094-RG179 Times AA3248 Teledyne 11299 Thermax 75-738-BCCWXE Tensolite 30888/L707YX-1 Haveg 8100207	12	D	B
417	Yellow	Brown	Violet	1/ M17/095-RG180 Raychem 9527D1514-2L Raychem 9528A1318 Microdot 293-3922			
418	Yellow	Brown	Gray	1/ Microdot 250-4070			
419	Yellow	Brown	White	1/ Raychem 48-502 & 5022E5111			
420	Yellow	Red	Black	1/ Raychem 48-950 & 9530D5117			
421	Yellow	Red	Brown	1/ Raychem 7624D1311 Raychem 9527A1318			
422	Yellow	Red	Red	1/ Gore GWN1159A			
423	Yellow	Red	Orange	1S50MU-16, -20, -40, -70 (MIL-C-915/42)			

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

Ⓐ TABLE II. Tools.

BIN code	Inner contact		Outer contact		Installing tool	Removal tool
	Basic crimping tool	Positioner	Basic crimping tool	Positioner		
416, 417, 418, 419, 420, 421, 422, 423	M22520/2-01	M22520/2-34	M22520/31-01	M22520/31-02	M81969/8-09 or M81969/14-04	M81969/8-10 or M81969/14-04

Ⓐ Cable to contact information: See table III.

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Ⓐ TABLE III. Cable to contact information.

BIN code	Cable accommodated	Inner contact tool selector setting no.
416	M17/119-RG174	4
	M17/113-RG316	5
	M17/094-RG179	3
	1/ 2/ Times AA3248	4
	Teledyne 11299	4
	Thermax 75-738-BCCWXE	4
	Tensolite 30888/L707YX-1	4
	Haveg 8100207	4
417	M17/095-RG180	3
	1/ Raychem 9527D1514-2L	5
	Raychem 9528A1318	4
	Microdot 293-3922	3
418	1/ Microdot 250-4070	4
419	1/ Raychem 48-502 & 5022E5111	4
420	1/ Raychem 48-950 & 9530D5117	3
421	1/ Raychem 7624D1311	5
	Raychem 9527A1318	5
422	1/ Gore GWN1159A	4
423	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	4

1/ Or equivalent
2/ High tensile strength copper alloy wire.

Contact resistance: See table IV.
Test current:
Inner contact - 1 ampere.
Outer contact - 12 amperes.

Low signal level contact resistance (inner contact only): See table V.

Contact engagement and separation forces (socket contact only): The engagement depth shall be as encountered in normal service. The test pins shall be in accordance with MS3197 except the diameters shall be as specified in the following, and surface roughness shall not exceed 3 micro-inches. Provision for clearance hole shall be provided.

Test pin diameter (inch)	Minimum separation force (ounces)		Maximum engagement force (ounces)		Maximum average engagement force
	Initial	After conditioning	Initial	After conditioning	
.0950 +.0002 -.0000	NA	NA	30	36	NA
.0930 +.0000 -.0002	3.0	2.5	NA	NA	NA

Dielectric withstanding voltage (applied between inner and outer contact):
Test voltage:
At sea level - 1000 Vac rms.
At 50,000 feet - 250 Vac rms.

Tensile strength (inner and outer contact crimp joint): See table V.

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TABLE IV. Contact resistance.

BIN code	Cable accommodated	Maximum voltage drop (millivolts) 25° +3° -0°C Inner contact	25° +3° -0°C Outer contact	2/ 25° +3° -0°C Inner contact	2/ 25° +3° -0°C Outer contact	200° +3° -0°C Inner contact	200° +3° -0°C Outer contact	Maximum average voltage drop
416	M17/119-RG174	55	85	66	102	94 4/	145 4/	NA
	M17/113-RG316	55	75	66	90	94	128	
	M17/094-RG179	120	70	144	84	204	119	
	1/3/ Times AA3248	170	150	204	180	290	255	
	Teledyne 11299	↓	↓	↓	↓	↓	↓	
	Thermax 75-738-BCCWXE							
	Tensolite 30888/L707YX-1							
	Haveg 8100207							
417	M17/095-RG180	120	60	144	72	204	102	
	1/ Raychem 9527D1514-2L	↓	↓	↓	↓	204 5/	102 5/	
	Raychem 9528A1318					204 5/	102 5/	
	Microdot 293-3922					204	102	
418	1/ Microdot 250-4070	↓		↓		204	102	
419	1/ Raychem 48-502 & 5022E5111	55		66		94 5/	102 5/	
420	1/ Raychem 48-950 & 9530D5117	120		144		204 5/	102 5/	
421	1/ Raychem 7624D1311	55		66		94 5/	102 5/	
	Raychem 9527A1318	120		144		204 5/	102 5/	
422	1/ Gore GWN1159A	120		144		204	102	
423	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	↓	66	↓	94 4/	102 4/	↓

1/ Or equivalent.
2/ After conditioning.
3/ High tensile strength copper alloy wire.
4/ 85° +3° -0°C.
5/ 115° +3° -0°C.

Ⓐ TABLE V. Low signal level contact resistance (inner contact only) and tensile strength.

BIN code	Cable accommodated	Maximum contact resistance (milliohms) Initial	Maximum contact resistance (milliohms) After conditioning	Tensile load (pounds) (minimum) Inner contact	Tensile load (pounds) (minimum) Outer contact
416	M17/119-RG174	55	66	15.0	15.0
	M17/113-RG316	55	66	10.0	15.0
	M17/094-RG179	120	144	3.5	15.0
	1/2/ Times AA3248	170	204	7.0	35.0
	Teledyne 11299	↓	↓	↓	↓
	Thermax 75-738-BCCWXE				
	Tensolite 30888/L707YX-1				
	Haveg 8100207				
417	M17/095-RG180	120	144	3.5	20.0
	1/ Raychem 9527D1514-2L	↓	↓	6.0	↓
	Raychem 9528A1318			6.0	
	Microdot 293-3922			3.5	
418	1/ Microdot 250-4070	↓	↓	4.0	
419	1/ Raychem 48-502 & 5022E5111	55	66	10.0	
420	1/ Raychem 48-950 & 9530D5117	120	144	3.5	
421	1/ Raychem 7624D1311	55	66	6.0	
	Raychem 9527A1318	120	144	6.0	
422	1/ Gore GWN1159A	120	144	4.0	
423	1S50MU-16, -20, -40, -70 (MIL-C-915/42)	55	66	15.0	↓

1/ Or equivalent.
2/ High tensile strength copper alloy wire.

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Ⓐ Vibration: Test condition VI, letter J, per method 2005 of MIL-STD-1344. Except directions shall be longitudinal and perpendicular. Time duration shall be 8 hours in each direction.

Ⓐ Shock (specified pulse): Test condition D, per method 2004 of MIL-STD-1344.

Ⓐ Shipping: Figure 1 illustrates final assembly; parts are to be shipped unassembled in a suitable container. Manufacturers recommended assembly instructions are to be shipped with the unit package.

QPL evaluating activity: Defense Electronics Supply Center (DESC-E), Dayton, OH 45444.

Military part number: See table VI..

Ⓐ TABLE VI. Military part number.

BIN code	Military part number
416	M39029/75-12A
417	M39029/75-12B
418	M39029/75-12C
419	M39029/75-12D
420	M39029/75-12E
421	M39029/75-12F
422	M39029/75-12G
423	M39029/75-12H

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR
Navy - EC, MC, OS
Air Force - 11, 15, 17, 99
DLA - ES

Preparing activity:
Air Force - 85

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